

Afghanistan (Multi-Grid Source) → Zahedan, Iran (Single Receptor)

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1) Objective

This model quantifies how much aerosol optical depth (AOD) over Afghanistan (AF) is transported toward Zahedan, Iran, under favorable wind conditions.

The analysis asks, for each month:

1. Which Afghan grid cells have wind vectors pointing toward Zahedan?
2. How much AOD is present in those upwind cells?
3. How far do those aerosols need to travel to reach Zahedan (distance decay)?
4. Does the resulting transport index co-vary with Zahedan's observed AOD?

This is a same-month, climatological analysis using 24-year monthly averages (2001–2024). It does not model individual dust storm events or time-lagged transport, but provides a physically grounded signal of the AF → Zahedan transport pathway.

Note on Zahedan AOD: Zahedan's observed AOD is used as a **validation reference only** — it is not an input to the WWTE model. The WWTE index itself

is the model’s estimate of transport contribution from Afghanistan. See Section 12 for full discussion.

Note on NDVI: NDVI (vegetation sink) has been intentionally removed from this pipeline. It will be incorporated in a separate vegetation analysis workflow and merged with these outputs at a later stage. See Section 16.

2) What Changed from the Old Code

The original script computed a transport index using only wind alignment and raw wind speed. Four significant physical and methodological improvements have been made in this version:

#	Component	Old Behavior	New Behavior
1	Distance decay	Not included — far and near pixels weighted equally	$\exp(-d/L)$ using geodesic Haversine distance
2	Wind direction geometry	Raw (d_{lon} , d_{lat}) degrees — flat-earth, distorted at mid-latitudes	$\cos(\text{lat})$ -corrected bearing — physically accurate at 30–40 degN
3	Angular threshold	Hard binary 30 deg cone — cliff edge at boundary	Soft cosine weighting — contribution fades smoothly with angle
4	Wind speed	Raw m/s — not comparable across months	Normalized to $[0, 1]$ within each month
5	Zahedan AOD	Single pixel lookup — fragile if grid misaligns	Area mean within 0.3 deg buffer — stable and representative
6	Index scaling	Raw unbounded values	Normalized to $[0, 1]$ across months for comparability

Why These Changes Matter

Distance decay is the most critical physical fix. Without it, an Afghan pixel 1,500 km from Zahedan receives the same weight as one 200 km away, even though aerosols deposit and dissipate during the long journey. The exponential decay $\exp(-d/L)$ reflects this physical reality.

$\cos(\text{lat})$ **correction** prevents directional bias. At 35 degN (Afghanistan’s latitude), one degree of longitude is only ~ 91 km while one degree of latitude is ~ 111 km. Using raw degree differences distorts the computed angle between wind and the pixel-to-Zahedan direction.

Soft cosine weighting replaces the old hard 30 deg angular cone. Previously, a pixel at 29.9 deg off-target received full weight while a pixel at 30.1 deg received zero — a physically unrealistic cliff edge. The cosine now smoothly weights contribution by how well-aligned the wind is.

3) Input Data

Data directory: /Users/hossein.lotfi/temp/NDVI/data/gdrive

3.1 Required Files

File Pattern	Variable	Resolution	Notes
Avg_AOD_24yr_Month{MM}_2001-2024.tif	Depth	~0.01 deg	24-year monthly mean
speed_{MM}.tif	Wind speed (m/s)	~0.1 deg	Scalar magnitude
u_avg_{MM}.tif	Zonal wind component	~0.1 deg	Positive = eastward
v_avg_{MM}.tif	Meridional wind component	~0.1 deg	Positive = northward

Replace {MM} with the two-digit month number (e.g., 06, 07).

NDVI files are **not required** in this pipeline. They will be used in a separate workflow (see Section 16).

3.2 Grid Details

- **AOD grid:** EPSG:4326, approximately 0.01 deg resolution
- **Wind grids (speed, u, v):** EPSG:4326, approximately 0.1 deg resolution
- **Resampling:** AOD is bilinearly resampled to the wind grid before any pixel-level calculation

3.3 Receptor Point

Parameter	Value
Location	Zahedan, Iran (city center)
Longitude	60.86 deg E
Latitude	29.50 deg N
AOD sampling	Area mean within 0.3 deg radius (~33 km buffer)

Why a buffer instead of a single pixel? A single-pixel lookup is fragile — grid misalignment or a cloud gap can silently return NaN or an unrepresentative value. Averaging all pixels within 0.3 deg (~33 km) of the city center gives a stable, robust reading. Importantly, this value is used only for validation, not as a model input.

4) Core Concept in Plain English

For each pixel inside Afghanistan, the model asks three questions:

1. **Direction:** Is the wind blowing toward Zahedan, and how well-aligned is it?
2. **Strength:** How fast is the wind, and how much AOD is at this pixel?
3. **Distance:** How far does the aerosol need to travel before reaching Zahedan?

Pixels that score well on all three questions contribute more to the transport index.

Analogy: Think of each Afghan pixel as a factory chimney. The WWTE score for that chimney depends on: - How much smoke it is producing (AOD) - Whether the wind is blowing the smoke toward Zahedan (alignment) - How hard the wind is blowing (speed) - How far Zahedan is — smoke thins out over long distances (distance decay)

5) Mathematical Formulation

5.1 Receptor Target Vector

For each Afghan pixel i at coordinates (lon_i, lat_i) , define the direction vector pointing **from pixel i toward Zahedan**, with a $\cos(lat)$ correction:

$$\Delta lon_i = (lon_{Zahedan} - lon_i) \times \cos\left(\frac{lat_i + lat_{Zahedan}}{2}\right)$$

$$\Delta lat_i = lat_{Zahedan} - lat_i$$

Why multiply Δlon by $\cos(lat)$? At 35 degN (roughly Afghanistan's latitude), one degree of longitude equals only ~91 km, while one degree of latitude equals ~111 km. Without this correction, east-west distances are underestimated and the wind alignment angle is physically distorted. The $\cos(lat)$ factor places longitude and latitude differences on the same physical scale.

Why Zahedan-minus-pixel (not pixel-minus-Zahedan)?

Example — pixel at (50E, 25N), Zahedan at (60.86E, 29.50N):

- $\Delta lon = 60.86 - 50 = +10.86 \rightarrow$ Zahedan is to the **east** ✓
- $\Delta lat = 29.50 - 25 = +4.50 \rightarrow$ Zahedan is to the **north** ✓

Reversing the subtraction produces a vector pointing **away** from Zahedan, causing the dot product to be positive for winds blowing in the wrong direction.

5.2 Wind Vector and Magnitudes

Wind vector at pixel i :

$$\mathbf{w}_i = (u_i, v_i)$$

Wind speed magnitude:

$$\|\mathbf{w}_i\| = \sqrt{u_i^2 + v_i^2}$$

Target direction magnitude (distance in corrected degree-space):

$$\|\mathbf{d}_i\| = \sqrt{\Delta lon_i^2 + \Delta lat_i^2}$$

Both magnitudes normalize the vectors before the alignment angle is computed. Only direction matters for alignment — not the magnitude of either vector.

5.3 Wind Alignment (Soft Cosine)

The cosine of the angle θ_i between the wind vector and the pixel-to-Zahedan direction:

$$\cos(\theta_i) = \frac{\mathbf{w}_i \cdot \mathbf{d}_i}{\|\mathbf{w}_i\| \|\mathbf{d}_i\|} = \frac{u_i \Delta lon_i + v_i \Delta lat_i}{\|\mathbf{w}_i\| \|\mathbf{d}_i\|}$$

$\cos(\theta_i)$	Angle	Meaning
1.00	0 deg	Wind blows perfectly toward Zahedan
0.87	30 deg	Wind is mostly toward Zahedan
0.50	60 deg	Partial alignment
0.00	90 deg	Wind is perpendicular — no transport
< 0	> 90 deg	Wind points away — pixel excluded

The old code applied a hard 30 deg binary cutoff. The new code uses $\max(\cos \theta_i, 0)$ as a continuous smooth weight — there is no cliff edge.

5.4 Normalized Wind Speed

$$\text{speed_norm}_i = \frac{\|\mathbf{w}_i\|}{\max_j \|\mathbf{w}_j\|}$$

Dividing by the monthly maximum yields a value in $[0, 1]$, making wind speed contributions comparable across months regardless of seasonal wind regime.

6) Distance Decay

Aerosols deposit, settle, and wash out as they travel. The exponential decay function models this loss over distance:

$$\text{decay}_i = \exp\left(\frac{-d_i}{L}\right)$$

Where: - d_i = geodesic (great-circle) distance in km, computed via the **Haversine formula** -
 L = decay length scale (DECAY_LENGTH_KM, default: **800 km**)

Distance to Zahedan	$\exp(-d/800)$	Signal Retained
200 km	0.78	78%
400 km	0.61	61%
800 km	0.37	37% (1 e-folding length)
1,200 km	0.22	22%
1,600 km	0.14	14%

Why Haversine instead of degree distance?

The old code used $\sqrt{\Delta lon^2 + \Delta lat^2}$ in raw degrees as a proxy for distance. This is physically incorrect because degrees of longitude shrink toward the poles. The Haversine formula computes the true great-circle distance in km:

$$d_i = 2R \arcsin \sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_i \cos\phi_{Zah} \sin^2\left(\frac{\Delta\lambda}{2}\right)}$$

where $R = 6371$ km, ϕ denotes latitude in radians, and λ longitude in radians.

Tuning guidance: Typical values range from 500 km (short-lived coarse particles) to 1,500 km (long-range fine aerosols). Start with 800 km and run sensitivity tests (see Section 13).

7) Full WWTE Score Formula

Combining all three physical components, the WWTE score for each source pixel i :

$$\text{score}_i = \underbrace{AOD_i}_{\text{source strength}} \times \underbrace{\max(\cos\theta_i, 0)}_{\text{wind alignment}} \times \underbrace{\text{speed_norm}_i}_{\text{wind speed [0,1]}} \times \underbrace{\exp\left(\frac{-d_i}{L}\right)}_{\text{distance decay}}$$

7.1 Aggregation Weight

Each pixel's contribution is weighted by its transport intensity (alignment x speed):

$$\omega_i = \max(\cos\theta_i, 0) \times \text{speed_norm}_i$$

7.2 WWTE Index

$$I_{\text{WWTE}} = \frac{\sum_i \text{score}_i \times \omega_i}{\sum_i \omega_i}$$

7.3 Worked Numerical Example

Three directional Afghan pixels in one month:

Pixel	AOD_i	$\cos \theta_i$	spd_norm_i	d_i (km)	$\exp(-d/800)$	score_i	ω_i	$\text{score}_i \times \omega_i$
A	1.5	0.90	0.85	350	0.644	0.741	0.765	0.567
B	0.8	0.65	0.60	700	0.415	0.130	0.390	0.051
C	2.2	0.55	0.95	250	0.732	0.842	0.523	0.440

$$I_{\text{WWTE}} = \frac{0.567 + 0.051 + 0.440}{0.765 + 0.390 + 0.523} = \frac{1.058}{1.678} \approx 0.631$$

Pixel A has excellent alignment and reasonable AOD. Pixel C has the highest AOD and is closest to Zahedan, so despite moderate alignment it scores well. Pixel B is far, moderately aligned, and has low AOD — its contribution is small.

8) Processing Steps

- ```

Step 1 --- Load wind grids
 Read speed_{MM}.tif, u_avg_{MM}.tif, v_avg_{MM}.tif
 Extract transform, CRS, and grid dimensions

Step 2 --- Load and resample AOD
 Read Avg_AOD_24yr_Month{MM}_2001_2024.tif (~0.01 deg)
 Bilinear resample to wind grid (~0.1 deg)

Step 3 --- Build Afghanistan mask
 Download Natural Earth 110m country boundaries
 Rasterize AFG polygon onto wind grid
 Result: boolean array (True = inside Afghanistan)

Step 4 --- Compute pixel coordinates
 Calculate lon/lat of each wind-grid pixel center

Step 5 --- Compute cos(lat)-corrected bearing to Zahedan [FIXED]
 Scale Deltalon by cos(mean_lat) before computing unit direction vector

Step 6 --- Compute geodesic distances (Haversine) [NEW]
 Great-circle km from each AF pixel to Zahedan (60.86 degE, 29.50 degN)

```

```

Step 7 --- Compute wind alignment (soft cosine) [FIXED]
 cos(theta) = dot(wind_unit_vector, target_unit_vector)
 No hard angular threshold --- smooth weighting only

Step 8 --- Normalize wind speed [FIXED]
 speed_norm = speed / max(speed within month)

Step 9 --- Compute distance decay [NEW]
 decay = exp(-distance_km / DECAY_LENGTH_KM)

Step 10 --- Compute per-pixel WWTE scores
 score = AOD x max(cos theta, 0) x speed_norm x decay

Step 11 --- Compute WWTE index
 I_WWTE = sum(score x omega) / sum(omega), where omega = max(cos theta, 0) x speed

Step 12 --- Compute Zahedan AOD (validation only) [FIXED]
 Area mean within 0.3 deg buffer --- not a model input

Step 13 --- Normalize WWTE index to [0, 1] across months [NEW]

Step 14 --- Write CSV and print summary

```

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## 9) How to Run

### 9.1 Prerequisites

```
[] pip install numpy pandas rasterio geopandas
```

### 9.2 File Setup

Ensure all required .tif files are in DATA\_DIR:

```

/Users/hosseini.lotfi/temp/NDVI/data/gdrive/
|-- Avg_AOD_24yr_Month06_2001_2024.tif
|-- Avg_AOD_24yr_Month07_2001_2024.tif
|-- speed_06.tif speed_07.tif
|-- u_avg_06.tif u_avg_07.tif
\-- v_avg_06.tif v_avg_07.tif

```

Note: NDVI files are NOT required for this pipeline.

### 9.3 Run the Script



```
[] cd /Users/hossein.lotfi/temp/NDVI python analyze_aod_af_zahedan.py
```

## 9.4 Output Location

```
/Users/hossein.lotfi/temp/NDVI/data/gdrive/analysis_out/
\-- af_zahedan_wwte_summary.csv
```

## 10) Output Files and Columns

Output CSV: af\_zahedan\_wwte\_summary.csv

| Column               | Description                                                                   |
|----------------------|-------------------------------------------------------------------------------|
| month                | Two-digit month string (e.g., "06")                                           |
| zahedan_aod          | Area-mean AOD at Zahedan within 0.3 deg buffer ( <i>validation only</i> )     |
| af_aod_mean          | Mean AOD across all valid Afghan pixels                                       |
| af_aod_toward_mean   | Mean AOD in pixels where wind points toward Zahedan                           |
| wwte_index           | Raw WWTE transport index:<br>$\sum(\text{score} \times \omega) / \sum \omega$ |
| wwte_index_norm      | WWTE index normalized to $[0, 1]$ across all months                           |
| toward_pixel_count   | Number of AF pixels with $\cos \theta > 0$ toward Zahedan                     |
| af_valid_pixel_count | Total number of valid (non-NaN) AF pixels                                     |
| mean_dist_decay      | Mean $\exp(-d/L)$ across directional pixels                                   |
| mean_cosang_toward   | Mean $\cos \theta$ across directional pixels                                  |
| high_toward_aod_mean | Mean AOD of the top-quartile directional pixels                               |
| high_toward_fraction | Fraction of directional pixels in the top AOD quartile                        |

## 11) Results from 2-Month Data (June & July)

### 11.1 Key Metrics (from old transport index — new WWTE values will differ)

| Metric                           | June   | July   | Change |
|----------------------------------|--------|--------|--------|
| Zahedan AOD                      | 1.3659 | 1.4016 | +2.6%  |
| AF mean AOD (all pixels)         | 1.3763 | 2.3124 | +68%   |
| AF AOD (directional pixels only) | 0.6935 | 0.7541 | +8.6%  |
| Old transport index              | 0.6946 | 0.7601 | +9.4%  |
| Directional pixel count          | 1,170  | 1,273  | +8.8%  |

| Metric                               | June  | July  | Change  |
|--------------------------------------|-------|-------|---------|
| Fraction of AF pixels toward Zahedan | 18.4% | 20.1% | +1.7 pp |

**Note:** New WWTE index values will differ from the old `transport_aod_index` because distance decay is now included. Re-run the script to obtain updated values.

## 11.2 Notable Observations

**Afghanistan AOD spikes in July but most of it does not go toward Zahedan.** The whole-domain AF AOD jumps +68% from June to July, but AOD in pixels actually pointing toward Zahedan rises only +8.6%. Most of July’s Afghan dust surge is directed elsewhere.

**July wind shifts slightly to favor Zahedan transport.** About 1,170 AF pixels (18.4%) pointed toward Zahedan in June; this rose to 1,273 pixels (20.1%) in July — a modest seasonal shift toward westward transport.

**Both the transport index and Zahedan AOD increase together.** This directional co-movement supports the AF → Zahedan transport hypothesis, though two months are insufficient for statistical inference.

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## 12) Interpretation

### 12.1 Role of Zahedan AOD in This Analysis

The Zahedan AOD value in the output is a **validation metric**, not a model input. The WWTE index is the model’s own computed estimate of how much Afghan aerosol is being transported toward Zahedan. Comparing WWTE against observed Zahedan AOD tests whether the index captures a real physical signal.

The WWTE model is self-contained: it estimates transport contribution using only Afghan source data (AOD, wind, distance). Zahedan observations are only needed if you want to evaluate model skill.

### 12.2 Why Is the Transport Index Lower Than Zahedan AOD?

The directional AF AOD subset ( $\sim 0.69$ – $0.75$ ) is consistently lower than Zahedan AOD ( $\sim 1.37$ – $1.40$ ). This is expected because:

1. The directional AF pixels are a geographic **subset** of all Afghanistan and may not include the highest-AOD regions.
2. Zahedan receives aerosol from **multiple sources** beyond Afghanistan: local emissions, the Persian Gulf, Pakistan, and other regional pathways.
3. Distance decay reduces the effective contribution during the long journey.

### 12.3 Correlation Warning

With only two months, any computed correlation equals  $r = 1.0$  by mathematical necessity and is statistically meaningless. **Do not report this as evidence of a relationship.** A meaningful correlation requires a minimum of 8–12 months.

### 13) Limitations and Caveats

| # | Limitation                          | Impact                                      | Mitigation                                          |
|---|-------------------------------------|---------------------------------------------|-----------------------------------------------------|
| 1 | Only 2 months analyzed              | No statistical inference possible           | Extend to all 12 months                             |
| 2 | 24-year monthly average data        | Daily/synoptic variability is smoothed      | Use daily data for event-level analysis             |
| 3 | No time lag modeled                 | Transport takes hours to days, not captured | Add 1–5 day lagged analysis                         |
| 4 | Single decay length ( $L = 800$ km) | May not match actual aerosol lifetime       | Sensitivity test with $L \in \{500, 800, 1200\}$ km |
| 5 | NDVI excluded                       | Vegetation sink not accounted for           | Will be added in separate workflow                  |
| 6 | No atmospheric chemistry            | Aerosol transformation not modeled          | Out of scope; use CTM for this                      |
| 7 | Zahedan has multiple sources        | AF is one contributor among several         | Compare with source apportionment                   |

### 14) Recommended Next Steps

#### Priority 1 — More Months

Add all 12 months (minimum: months 05–10, peak dust season) to establish a seasonal pattern and compute a statistically meaningful correlation ( $n \geq 10$ ).

#### Priority 2 — Sensitivity Analysis on Decay Length

```
[] DECAY_LENGTH_KM = [500, 800, 1200] # test all three
```

Report how much  $I_{\text{WWTE}}$  changes across these values to assess model sensitivity.

#### Priority 3 — NDVI Vegetation Workflow

Develop the separate NDVI workflow (see Section 16) and merge its output with the WWTE index to produce a fully physics-informed transport score.

## Priority 4 — Bootstrap Confidence Intervals

Once enough months are available, apply bootstrap resampling to compute 95% confidence intervals around  $I_{\text{WWTE}}$  and its correlation with Zahedan AOD.

## Priority 5 — Daily Data

Repeat with daily AOD and wind fields to capture individual transport episodes (dust storms, smoke events) rather than climatological averages.

## Priority 6 — Direction Decomposition

Split Afghanistan into sub-regions (southern, eastern, northern, western) to identify which part contributes most to Zahedan transport.

## Priority 7 — Validation Against Independent Data

Compare the WWTE index against: - HYSPLIT back-trajectory cluster frequencies initialized at Zahedan - CALIPSO aerosol vertical profiles along the AF–Zahedan corridor - Surface PM<sub>10</sub> station observations at Zahedan (if available)

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## 15) Key Parameters Reference

| Parameter          | Variable Name        | Default     | Tuning Range | Effect                                   |
|--------------------|----------------------|-------------|--------------|------------------------------------------|
| Decay length $L$   | DECAY_LENGTH_KM      | 800 km      | 500–1,500 km | Higher → more weight to distant pixels   |
| Zahedan buffer     | ZAHEDAN_BUFFER_DEG   | 0.3 deg     | 0.1–0.5 deg  | Larger → more pixels in area mean        |
| Receptor longitude | ZAHEDAN_LON          | 60.86 deg E | —            | City center coordinate                   |
| Receptor latitude  | ZAHEDAN_LAT          | 29.50 deg N | —            | City center coordinate                   |
| High-AOD quantile  | AF_HIGH_AOD_QUANTILE | 0.5         | 0.5–0.9      | Threshold for high-AOD diagnostic metric |

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## 16) Future: NDVI Vegetation Workflow

NDVI has been intentionally excluded from this pipeline so it can be developed and validated independently. When ready, the NDVI modifier will be integrated as an additional multiplicative term in the WWTE score:

$$\text{score}_i^{\text{full}} = \text{score}_i^{\text{WWTE}} \times \underbrace{(1 - \alpha \cdot \text{NDVI}_i)}_{\text{vegetation sink}}$$

Where  $\alpha \in [0, 1]$  (default 0.5) controls how strongly vegetation absorbs aerosols during transport.

| Land Cover        | Typical <i>NDVI</i> | Veg Sink Factor ( $\alpha = 0.5$ )                        |
|-------------------|---------------------|-----------------------------------------------------------|
| Bare desert       | 0.05                | $1 - 0.5 \times 0.05 = 0.975$ —<br>minimal absorption     |
| Sparse scrubland  | 0.30                | $1 - 0.5 \times 0.30 = 0.850$ —<br>moderate absorption    |
| Agricultural land | 0.55                | $1 - 0.5 \times 0.55 = 0.725$ —<br>significant absorption |
| Dense vegetation  | 0.75                | $1 - 0.5 \times 0.75 = 0.625$ —<br>strong absorption      |

### Required file for future integration:

NDVI\_Month{MM}.tif      (one file per month, resampled to 0.1 deg wind grid)

The NDVI workflow will produce a per-pixel vegetation sink map that can be directly multiplied into the existing WWTE score arrays without restructuring the current pipeline.

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*Model: WWTE v2.1 — Wind-Weighted AOD Transport Efficiency (NDVI-free)*

*Study region: Afghanistan → Zahedan, Iran*

*Data period: 2001–2024 (24-year monthly averages)*

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